Docker Image and Container Build

Step#1 Goto Bios and enable virtualization

Step#2 Goto Turn Windows Features on window and checkbox of hiper-v and windows subsystem for linux

Step#3 Goto Microsoft app store and install Ubuntu

Step#4 Install Docker-Desktop and Integrate wsl Ubuntu in the Settings>>General and Resource portion

Step#5 Install VScode and add extention of wsl, docker and python

Step#6 Create Flask API and requirements.txt

Note: Create requirements.txt:

Command: pip freeze >requirements.txt

Step#7 Create a Dockerfile:

Note: In The Dockerfile:

```
# Use an official Python runtime as a parent image
FROM python:3.11.9
# Set the working directory in the container
WORKDIR /abc
# Copy the current directory contents into the container at /app
COPY . /abc
# Install any needed packages specified in requirements.txt
RUN pip install --no-cache-dir -r requirements.txt
# Make port 80 available to the world outside this container
EXPOSE 80
# Define environment variable
ENV NAME World
# Run app.py when the container launches
CMD ["python", "abc.py"]
```

Step#8 Goto wsl Terminal To Build a Docker Image:

Note: If Need to Upgrade pip then run the command: sudo apt install python3-pip

Command: ubuntuuser$ docker build -t abc .

Command: ubuntuuser$ docker images Note: Check the images

Step#9 Build Docker Container:

Command: ubuntuuser$ docker run -p 4000:80 abc

Step#10 Check the docker container status:

Command: ubuntuuser$ docker ps

Step#11 Goto Browser and enter the url: localhost:3000

Step#12 To Stop Container:

Command: ubuntuuser$ docker stop 411bfdb13f93 Note: Container ID: 411bfdb13f93

Command: ubuntuuser$ docker start 411bfdb13f93

Step#13 Check The Log of Container:

Command: ubuntuuser$ docker logs 411bfdb13f93 Note: Container ID or Name

Command: ubuntuuser$ docker logs -f 411bfdb13f93 Note: Print Continuous Log

Command: ubuntuuser$ docker logs -f --tail 10 411bfdb13f93 Note: Print Last 10 line of Log

Step#14 Goto hub.docker.com and login as docker user then find the pushed container for anyone use

Step#15 Save Docker Image in a Local Machine:

Command: C:\Users\me>docker save --output="Desktop\aaa.tar" abc Note: aaa.tar is a new tar file and 'wow' is a image name

Step#16 Load Docker Image:

Command: C:\Users\me>docker load --input="Desktop\aaa.tar"

Step#17 Remove Docker Image:

Command: ubuntuuser$ docker rmi 5bf6a2e3bc31

Command: ubuntuuser$ docker rmi -f 5bf6a2e3bc31 Note: Forcedly Remove Single or Multiple Image

Step#18 Push The Docker Container in The Docker Hub:

Command: ubuntuuser$ docker push abc

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